Hyperspectral Band Selection for Zinc Concentrate Composition Prediction via Bayesian Nonparametric Segmentation and Multiobjective Optimization

Yan Sun, Can Zhou, Xueqian Yu, and Xuheng Zhan

***Abstract*—Hyperspectral imaging (HSI) provides a rapid and nondestructive technique for compositional analysis of zinc concentrate; however, the intrinsic high dimensionality of spectral data often results in significant redundancy, complicating subsequent modeling and analysis. Clustering-based band selection (BS) methods have shown promise in reducing dimensionality while retaining informative bands, yet these approaches fail to maintain spectral contiguity, adequately capture interband similarity, and optimize the quality of selected subsets comprehensively. To overcome these challenges, we propose a Bayesian nonparametric segmentation framework integrated with evolutionary multiobjective optimization for robust and efficient BS. Initially, kernel density estimation (KDE) is adopted to characterize the reflectance distribution of each spectral band. Bayesian inference then quantifies interband similarity and groups adjacent bands within a maximum-posterior framework, thus preserving the inherent wavelength contiguity and physical interpretability. To ensure computational efficiency, the band-grouping problem is transformed into a dynamic-programming formulation and resolved by employing the pruned exact linear time (PELT) algorithm, whose overall computational complexity is $O(n)$ with respect to the number of spectral bands. Subsequently, representative BS is modeled as a multiobjective optimization problem, balancing redundancy minimization and information maximization, integrating prior knowledge of element-sensitive wavelengths. Moreover, an innovative integer encoding strategy is introduced to significantly reduce computational overhead relative to traditional binary encoding methods. Finally, the proposed Bayesian nonparametric segmentation with multiobjective optimization (BNS-MOP) algorithm is validated against ten state-of-the-art band-selection techniques on hyperspectral data from zinc-concentrate samples. Across four target elements (Zn, Fe, Pb, and Cu), BNS-MOP consistently achieved superior performance, confirming the practical efficacy of the redundancy-aware, information-maximizing approach.**

***Index Terms*—Dynamic programming, hyperspectral band selection (BS), kernel density estimation (KDE), zinc concentrate composition prediction.**

I. Introduction

ZINC concentrate is a critical raw material in zinc smelting, primarily comprising zinc-bearing minerals, such as sphalerite (ZnS), smithsonite ($ZnCO_3$), marmatite ($Zn_xFe_{1-x}S$), and willemite (Zn_2SiO_4), along with impurities like iron, lead, copper, and sulfur [1]. Due to varying sources, zinc concentrate composition significantly fluctuates, necessitating precise compositional analysis before smelting for stable operations and enhanced product quality. Traditional analysis methods, such as chemical titration, although accurate, involve extensive sample preparation and prolonged processing times [2]. In contrast, hyperspectral imaging (HSI) offers rapid, nondestructive compositional analysis with high spatial and spectral resolution, efficiently characterizing heterogeneous mineral samples [3]. However, certain spectral bands exhibit weak or negligible correlations with compositional variations, resulting in redundant spectral features and information overlap. In addition, spectral reflectance across different bands is variably influenced by external noise, complicating computational modeling, increasing processing complexity, and reducing prediction accuracy due to the well-known Hughes phenomenon. Band selection (BS) techniques have emerged to address these challenges by reducing spectral redundancy while preserving critical information [4].

BS methods include supervised [5], semi-supervised [6], and unsupervised approaches [7]. Supervised methods select informative bands with the guidance of labeled samples, while semi-supervised methods utilize both a small amount of labeled data and a larger amount of unlabeled data. However, in practical zinc smelting processes, collecting sufficient and representative labeled samples is often costly and labor-intensive due to complex sampling and analysis procedures. In contrast, unsupervised BS methods do not require labeled data. Instead, they select bands by exploiting intrinsic spectral properties, such as redundancy and information content [8]. This makes unsupervised approaches suitable for industrial applications, where labeled datasets are limited or unavailable.

Unsupervised BS methods are commonly categorized into four kinds: ranking-based [9], sparse-representation-based [10], searching-based [11], and clustering-based [12], each employing distinct selection strategies. Ranking-based methods evaluate individual spectral bands according to predefined criteria and select those deemed most important [13]. Common ranking measures are typically based on information-theoretic metrics, such as the self-mutual information-based BS approach [14], local density- and information entropy-based BS method based on the neighborhood grouping normalized

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matched filter (NGNMF) [15], information divergence-based methods [16], [17]. Although this approach effectively identifies discriminative bands, it may introduce redundancy in the selected subset. Sparse-representation-based methods leverage the principle that each band can be reconstructed from a sparse combination of other bands, formulating the problem as an optimization under sparsity constraints [18]. While this framework can enhance interpretability and mitigate redundancy, it often incurs substantial computational cost due to the large number of hyperspectral pixels, and the final performance strongly depends on the chosen sparse representation model. Searching-based methods treat BS as a combinatorial optimization problem, employing greedy or evolutionary algorithms to identify optimal or near-optimal subsets [19]. Despite their effectiveness, they face considerable computational challenges owing to the large search space and the nonlinear nature of the optimization process. Clustering-based methods, which group bands based on similarity, have shown superior performance and, thus gained widespread adoption in recent years [20]. Clustering-based BS methods generally involve two main steps: 1) quantifying similarity among hyperspectral bands and grouping them into clusters to maximize intracluster similarity and minimize intercluster similarity, and 2) evaluating the performance of individual bands to select representative bands from each cluster, forming the final subset. Current studies mainly focus on improving these two aspects.

To quantify similarity and group hyperspectral bands, several clustering-based approaches have been proposed in recent studies. Sun et al. [21] proposed a hyperbolic clustering-based band hierarchy for unsupervised hyperspectral BS. A hyperbolic distance-based cluster-center ranking strategy is employed to weight the significance of each band and locate the center of each cluster, which can directly output a better BS result during the clustering process. In [22], a sparse deep subspace clustering method that captures the nonlinear subspace structure of hyperspectral data using a novel sparsity measure was proposed. Bands were clustered accordingly, and representative bands were selected using a multicriteria approach. Whereas hyperspectral bands are often treated as unordered, yet they inherently follow a wavelength-based sequence. Clustering spectrally similar but physically distant bands disrupts this contiguity, compromising physical interpretability and potentially introducing selection bias. Furthermore, disregarding this natural order requires additional post-processing to extract clustering indicators, increasing computational complexity. To address the challenge of achieving a reasonable clustering of hyperspectral bands while preserving their inherent wavelength contiguity, several segmenting-based approaches have been proposed. Wang et al. [23] introduced a dynamic-programming-based clustering framework that optimizes band partitioning by evaluating and maximizing a global clustering quality measure. Despite the NP-hard nature of the problem for many objective functions, their approach can still determine optimal partitions under reasonable constraints. To mitigate the issue of grouping spectrally similar but nonadjacent bands, Li et al. [24] proposed an intragroup similarity-based method that partitions hyperspectral bands while maintaining their original order. Unlike conventional approaches, this method defines intragroup similarity using all bands within a group, ensuring that selected bands are more dispersed. Another approach, proposed by Li et al. [25], leverages the block-diagonal structure of the spectral similarity matrix. By applying graph clustering with L_2-norm-based partitioning, hyperspectral bands are grouped into ordered clusters, allowing the similarity matrix to be directly used in subsequent analysis without requiring additional clustering indicators. Despite these advancements, most existing clustering-based methods rely on simplified similarity metrics such as Euclidean distance or correlation measures computed from representative statistics. However, spectral similarity between hyperspectral bands is inherently determined by their entire spectral distributions rather than merely certain summary statistics. Two bands may exhibit similar mean values or small Euclidean distances yet differ significantly in higher order distribution characteristics, potentially leading to suboptimal or physically inconsistent clustering results. Therefore, there remains a critical need for more comprehensive similarity measures that fully incorporate the underlying spectral distribution information of hyperspectral bands, which could further improve the accuracy and interpretability of clustering-based BS methods.

To effectively select representative bands within each cluster, several strategies have been developed in recent studies. The most straightforward approach is to select bands closest to cluster centroids. For example, Sun et al. [26] proposed a fast, latent low-rank subspace clustering framework that employs correntropy to measure spectral similarity between bands, followed by spectral clustering based on a sparse affinity matrix. Representative bands are then selected from each cluster by proximity to the cluster centroid. Similarly, Huang et al. [27] proposed a tensor-based clustering approach, grouping bands in a latent low-dimensional feature space, and selecting the bands nearest to the cluster centroids. Several ranking-based strategies were proposed to further improve BS performance. Jia et al. [28] combined clustering and ranking by first adaptively dividing bands into groups and subsequently performing clustering. Representative bands were selected via intragroup ranking. In [29], a graph-regularized spatial–spectral clustering approach was proposed, in which bands with the highest information entropy within each cluster were selected as representatives. In recent years, density-based criteria have shown significant promise for hyperspectral BS. In [30], the enhanced fast density-peak-based clustering (E-FDPC) algorithm clusters spectral bands and ranks them by a weighted combination of normalized local density and intracluster distance. An exponential learning rule is also used to adaptively set the cutoff threshold based on the desired number of bands. Chang et al. [31] introduced band density prominence clustering-based BS (BDPC-BS) method, a novel algorithm motivated by the need to overcome the limitations of existing DPC-based BS methods, such as their failure to address interband correlation. The algorithm achieves this by introducing a new concept called band prominence value (BPV) as a third indicator, which is combined with band local density and band distance to derive new band prioritization

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criteria. Although existing methods have shown promising performance in BS tasks, most current approaches evaluate spectral bands independently in each band cluster, neglecting the optimization of the collective performance of the selected subset. Consequently, selected subsets may lack global spectral representativeness and discriminative power. To address this limitation, it is essential to develop appropriate strategies that consider redundancy reduction and global spectral information simultaneously.

Multiobjective evolutionary computation, known for its global search capability, has been widely explored for hyperspectral BS. Sun et al. [32] proposed MOBS-TD, a multiobjective optimization (MOO)-based BS algorithm for hyperspectral target detection that models BS as an MOO problem by designing three objectives, namely information content, redundancy, and a novel metric for target-background separability, which is subsequently solved using an enhanced multiobjective particle swarm optimization algorithm. Wang et al. [33] proposed a multiobjective evolutionary multitasking algorithm for hyperspectral BS aimed at image classification, which models BS as an MO problem designed to simultaneously optimize interclass separation distance and band distribution uniformity. In [34], an MO algorithm for unsupervised hyperspectral BS tailored for anomaly detection was proposed. It models the problem by minimizing objectives related to band deviation, signal-to-noise ratio estimation, and interband redundancy, solved via a multiobjective evolutionary approach featuring a novel MO band similarity ranking strategy. However, existing MO-based methods commonly formulate BS as a binary (0–2) optimization problem [35], resulting in high-dimensional decision spaces due to the large number of hyperspectral bands. This significantly increases computational complexity, making efficient encoding strategies crucial, as representation directly impacts the performance of MO-based BS.

According to the above, although clustering-based hyperspectral BS methods have shown promising performance, there are still several important limitations. One prevalent issue is the reliance of many clustering-based approaches on overly simplistic similarity metrics, which do not adequately capture the spectral distribution information. This often leads to suboptimal band grouping and potentially less reliable analytical outcomes. Another critical limitation is that existing BS methods frequently disrupt the inherent wavelength order of hyperspectral bands, compromising the physical interpretability of the selected subsets. Moreover, the majority of existing BS strategies evaluate candidate bands in isolation or independently within each cluster, neglecting the collective global representativeness and complementary information, thus failing to ensure optimal overall performance. To overcome these limitations, this article proposes a hyperspectral BS method integrating Bayesian nonparametric segmentation with multiobjective optimization (BNS-MOP). The primary contributions are as follows.

1) A Bayesian nonparametric segmentation framework is developed, where hyperspectral bands are grouped through Bayesian inference to maximize posterior probability. To efficiently solve this optimization problem, it is reformulated into a dynamic-programming model enhanced by the pruned exact linear time (PELT) algorithm. This approach preserves inherent wavelength contiguity, ensures physical interpretability, and eliminates dependence on predetermined cluster numbers.
2) A novel similarity measure is proposed based on kernel density estimation (KDE), which explicitly characterizes spectral similarity through the reflectance distribution of pixels within each hyperspectral band, thereby enhancing the accuracy and effectiveness of band grouping.
3) BS is explicitly formulated as an MOO problem that simultaneously balances redundancy minimization and information maximization of selected bands. Domain-specific knowledge of wavelength-sensitive bands associated with critical elements in zinc concentrate is incorporated into the objective functions to enhance the practical relevance and effectiveness of the selected bands.
4) An innovative integer encoding strategy is introduced for the MOO framework. Each integer corresponds directly to a representative band from a segmented group, substantially reducing the dimensionality of the decision space and improving computational performance relative to traditional binary encoding methods.

The remainder of this article is structured as follows. Preliminary knowledge is provided in Section II. Section III describes the proposed algorithm, offering a comprehensive explanation of its implementation and detailing the reasoning behind the developed strategies. Section IV presents the experimental results and analysis. Finally, Section V draws conclusions from the findings.

II. Preliminaries

A. Hyperspectral BS Problem

Let $\mathbf{H} \in \mathbb{R}^{N_1 \times N_2 \times L}$ be a hyperspectral image, where N_1 and N_2 indicate the spatial width and height, respectively, and L is the number of spectral bands. Suppose $\mathbf{H_s}$ is the selected band subset and K is the number of selected bands, with $K \ll L$. The common definition of hyperspectral BS is expressed as follows [36]:

$$\mathbf{H_s} = [h_1, h_2, \ldots, h_L]$$
$$h_i = \begin{cases} 0, & \text{The } i\text{th band is selected} \\ 1, & \text{The } i\text{th band is not selected.} \end{cases} \tag{1}$$

The task of hyperspectral BS is essentially a combinatorial optimization problem. There exists a total of C_L^K possible subsets. As L increases, the size of the search space grows dramatically, causing the commonly used binary (0–2) encoding to become inefficient and impractical. Therefore, it is necessary to develop a more efficient encoding strategy to better handle the growing complexity of this optimization problem.

B. Bayes' Theorem

Bayes' theorem provides a fundamental framework for updating probabilities based on observed evidence. Formally,

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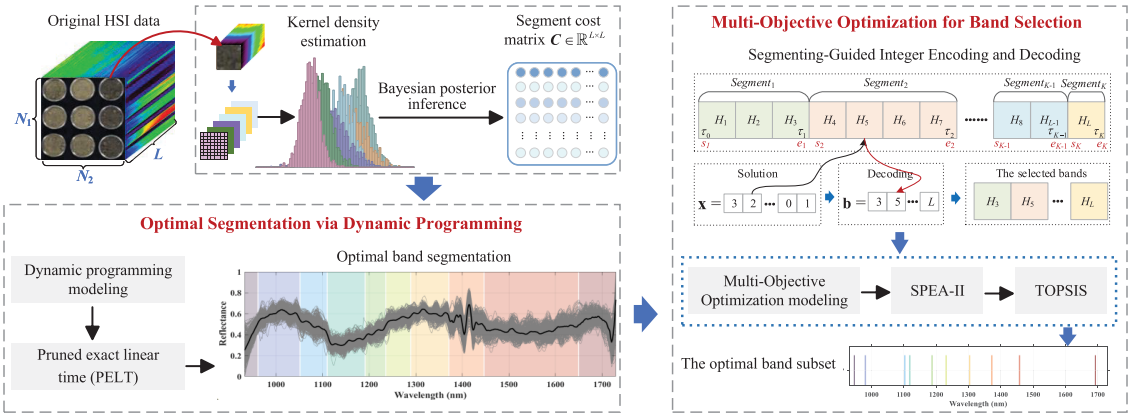


Fig. 1. Framework of BNS-MOP.

given events A and B, where $P(B) \neq 0$, Bayes' theorem can be stated as [37] and [38]

$$P(A|B) = \frac{P(B|A)\,P(A)}{P(B)} \tag{2}$$

where $P(A|B)$ represents the posterior probability of event A given event B, $P(B|A)$ is the likelihood of observing event B given event A, $P(A)$ is the prior probability of event A, and $P(B)$ is the marginal likelihood (or evidence) of event B. In Bayesian inference, this theorem is crucial for integrating prior knowledge with empirical data to yield updated posterior distributions.

III. Proposed Method

A. Overall Framework of Proposed Method

To address the challenge of redundant and high-dimensional hyperspectral data for zinc concentrate analysis, the proposed BNS-MOP algorithm operates in two tightly integrated phases: sequential band segmentation and band subset selection based on MOO. In particular, hyperspectral bands are initially partitioned using a Bayesian nonparametric segmentation strategy, where band similarity is quantified via KDE to explicitly capture reflectance distribution characteristics. After segmentation, the BS from each segment is formulated as an MOO problem that simultaneously considers redundancy minimization and information maximization. Solutions are encoded via a segmentwise integer strategy: each decision variable selects one band per segment and is decoded back to global band indices, drastically reducing the search dimensionality compared to binary encoding. The multiobjective search is conducted via strength Pareto evolutionary algorithm (SPEA-II), and the final band subset is chosen by a multiattribute decision-making method, thereby ensuring a compact set of K bands that are both minimally redundant and maximally informative. For a visualized display, Fig. 1 depicts the workflow of BNS-MOP. The detailed explanations are as follows.

B. Band Segmentation Based on Bayesian Nonparametric Modeling and Dynamic Programming

Numerous physical and empirical studies have shown that in the visible–near-infrared spectrum, absorption features such as those associated with soil minerals are typically much broader than the bandwidth of individual sensor channels. As a result, neighboring bands sample the same physical phenomena and inevitably carry redundant information [39]. This redundancy manifests as smooth transitions and strong correlations between adjacent wavelengths [40], [41]. Recent modeling approaches further support this view by grouping contiguous bands using shared the same Gaussian distribution components in hierarchical frameworks [42]. Motivated by these, we assume local stationarity within narrow spectral intervals and segment the spectrum into contiguous, locally homogeneous regions as the foundation for our Bayesian BS strategy.

1) Bayesian Modeling and Posterior Inference: The hyperspectral data $\mathbf{H}$ is transformed as a sequence of spectral bands denoted by $\mathbf{Y} = \{\mathbf{y}_1, \mathbf{y}_2, \ldots, \mathbf{y}_L\}$, where each $\mathbf{y}_\ell \in \mathbb{R}^N$ ($N = N_1 \times N_2$) corresponds to observations at the ℓth band. This article assumes that there exist K contiguous homogeneous segments along the spectral dimension, separated by unknown change points denoted as $\tau_1, \tau_2, \ldots, \tau_{K-1}$, satisfying

$$1 = \tau_0 < \tau_1 < \cdots < \tau_{K-1} < \tau_K = L. \tag{3}$$

Within each segment, the spectral observations are assumed to be independently drawn from an unknown probability distribution. Accordingly, for the jth segment (from band $\tau_{j-1}+1$ to band τ_j), the segmentwise likelihood function is given by

$$P_j = \prod_{\ell=\tau_{j-1}+1}^{\tau_j} \phi_j(\mathbf{y}_\ell) \tag{4}$$

where $\phi_j(\cdot)$ denotes the probability density function for the jth segment, and $\phi_j(\mathbf{y}_\ell)$ is formulated as the product of the kernel density estimates evaluated at all observed pixel values within the ℓth band

$$\phi_j(\mathbf{y}_\ell) = \prod_{q=1}^{N} \hat{\varphi}_j\left(y_{\ell,q}\right).$$

This multiplicative formulation of P_j encourages statistical homogeneity within each segment by assigning a higher likelihood to bands that are well represented by the estimated segment distribution. Conversely, bands that deviate from this distribution reduce the overall segment likelihood, thereby

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penalizing heterogeneous partitions. As a result, bands that are poorly explained by the segment distribution are less likely to be grouped within the same segment.

However, since the exact form of φ_j is unknown, we employ KDE [43], [44], also known as the Parzen window density estimation, to empirically approximate it from the observed data within each segment. The KDE method is a nonparametric technique. Unlike parametric methods, which assume a specific form of the underlying probability distribution, nonparametric methods do not make such assumptions and instead estimate the density directly from the observed data. KDE achieves this by centering a kernel function at each data point and averaging the contributions. This property allows KDE to flexibly adapt to the true underlying distribution, making it a classic nonparametric estimator. The KDE-based estimated density is then denoted as $\hat{\varphi}_j(y_{\ell,q})$. In particular, for scalar-valued $y_{\ell,q} \in \mathbb{R}$, the $\hat{\varphi}_j(y_{\ell,q})$ is formulated as

$$\hat{\varphi}_j(y) = \frac{1}{n_j h_j} \sum_{(\ell,q)\in\mathcal{V}_j} \Psi\left(\frac{y - y_{\ell,q}}{h_j}\right) \tag{5}$$

where $\mathcal{V}_j$ denotes the set of all pixel values across all bands in the jth segment: $\mathcal{V}_j = \{y_{\ell,q} \mid \tau_{j-1} < \ell \le \tau_j,\ 1 \le q \le N\}$. The sample size $n_j = N(\tau_j - \tau_{j-1})$. $\Psi(\cdot)$ is the kernel function, and h_j is the bandwidth.

As the kernel density estimator $\hat{\varphi}_j(y)$ is fully determined by the choice of kernel function and the bandwidth parameter h_j, it is critical to select these components appropriately to ensure reliable density estimation. In this study, the Gaussian kernel is adopted due to its smoothness, infinite support, and theoretical optimality properties. With the Gaussian kernel $\Psi(u) = (1/(2\pi)^{1/2})\exp(-(u^2/2))$, the density estimator becomes

$$\hat{\varphi}_j(y) = \frac{1}{n_j h_j} \sum_{(\ell,q)\in\mathcal{V}_j} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{y - y_{\ell,q}}{h}\right)^2\right). \tag{6}$$

Then, the bandwidth selection is based on minimizing the asymptotic mean integrated squared error (AMISE), a standard criterion that quantifies the overall estimation error. Following Silverman's rule of thumb, which yields near-optimal AMISE performance for Gaussian kernels, the bandwidth is determined as [45]:

$$h_j = 1.06\,\hat{\sigma}_j\, n_j^{-1/5} \tag{7}$$

where $\hat{\sigma}_j$ is the sample standard deviation of all entries in $\mathcal{V}_j$. This strategy ensures that the density estimator is both adaptive to local data variation and theoretically grounded in nonparametric estimation principles.

Why Parzen window instead of k-NN. The KDE is selected over alternatives such as k nearest neighbor density estimation to ensure robust segment costs and stable boundaries. In particular, KDE, also known as Parzen window estimation, provides two advantages. First, we apply the same bandwidth selection rule to every band. The rule maps each band's sample dispersion and its segment size n_j to a data-dependent bandwidth h_j, which keeps the amount of smoothing comparable across bands. Second, the estimation method itself guarantees smoothness. A KDE is constructed by placing a smooth, differentiable Gaussian kernel at each data point and summing them together. The resulting overall density estimate is therefore also a smooth, differentiable function, because the sum of differentiable functions is itself differentiable. This mathematical property is crucial, as it ensures that the resulting cost function $C(i, j)$ in the subsequent calculations is also robust to small perturbations in the data. In contrast, the radius in k nearest neighbor estimates changes with local density, leading to an inconsistent degree of smoothing across different bands. Moreover, the resulting estimates are continuous but mathematically nonsmooth. The estimated function has sharp corners where the neighbor set changes, which are points of nondifferentiability. These properties can make the cost function highly sensitive to minor data variations and lead to unstable segment boundaries. Such instability is detrimental because the min operation within succeeding dynamic-programming recursion is sensitive to small cost differences and can alter the optimal segmentation path returned, reducing the reliability and reproducibility of the results.

Assuming independence across segments, the joint likelihood function for the entire dataset $\mathbf{Y}$ given K segments and change points $\{\tau_j\}$ can be written as

$$P\left(\mathbf{Y} \mid K, \{\tau_j\}\right) = \prod_{j=1}^{K} \prod_{\ell=\tau_{j-1}+1}^{\tau_j} \hat{\phi}_j(\mathbf{y}_\ell). \tag{8}$$

To fully specify the Bayesian segmentation model, prior distributions for both the number of segments K and the locations of change points $\{\tau_j\}$ are required. Considering the absence of specific prior knowledge, noninformative priors are utilized in this article. In particular, a geometric prior, $P(K) \propto (0.5)^K$, is chosen for the segment number K, mildly penalizing excessive segmentations and, thus controlling model complexity. Conditioned on K, the change points $\{\tau_1, \tau_2, \ldots, \tau_{K-1}\}$ are uniformly distributed across all possible locations within the spectral dimension

$$P\left(\{\tau_j\} \mid K\right) = \frac{1}{\binom{L-1}{K-1}}. \tag{9}$$

By combining the likelihood function with these priors via Bayes' theorem, the joint posterior distribution for the segmentation parameters becomes

$$\begin{aligned} P\left(K, \{\tau_j\} \mid \mathbf{Y}\right) &\propto P\left(\mathbf{Y} \mid K, \{\tau_j\}\right) P\left(\{\tau_j\} \mid K\right) P(K) \\ &\propto \left[\prod_{j=1}^{K} \prod_{\ell=\tau_{j-1}+1}^{\tau_j} \hat{\phi}_j(\mathbf{y}_\ell)\right] \cdot \frac{1}{\binom{L-1}{K-1}} \cdot (0.5)^K. \end{aligned} \tag{10}$$

According to Bayesian inference, maximizing this joint posterior distribution yields the statistically optimal segmentation.

2) Construction of the Cost Function: The goal of maximum likelihood segmentation is to partition the spectral bands into contiguous, statistically homogeneous segments, with each segment modeled by a single probability distribution. In principle, this requires searching over all possible segmentation schemes. However, the number of possible ways to divide L bands into segments is 2^{L-1}, which makes brute-force optimization computationally infeasible even for moderate L. To address the computational challenge of segmentation, we adopt a dynamic programming approach that relies on

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the explicit precomputation of a segmentwise cost matrix. In particular, for each possible contiguous interval $[i, j]$, where i and j denote the start and end indices of a candidate segment ($1 \le i \le j \le L$), we calculate the modeling cost $\mathbf{C}(i, j)$. This cost is rigorously defined as the negative log-likelihood of all spectral bands within the interval $[i, j]$ being generated from a single underlying probability distribution. Formally, the cost of a segment spanning bands i to j is defined as

$$\mathbf{C}(i, j) = -\sum_{\ell=i}^{j} \log \hat{\phi}(\mathbf{y}_\ell). \tag{11}$$

The total number of such candidate segments is $O(L^2)$, making the cost matrix computation tractable even for moderately large L. Once this matrix is obtained, the optimal segmentation, defined as the set of segment boundaries that minimizes the total cost, can be efficiently determined using dynamic programming.

Consequently, the total segmentation cost, equivalent to the negative log-posterior, becomes

$$C_{\text{Total}} = \sum_{j=1}^{K} \mathbf{C}\left(\tau_{j-1} + 1, \tau_j\right) + \lambda(K - 1) \tag{12}$$

where $\lambda(K-1)$ is introduced as a penalty term to avoid excessive segmentation, and $\lambda > 0$ is a penalty parameter regulating the tradeoff between segmentation precision and the number of segments. In practice, λ can be set within the range between the minimum and the median of all candidate segment costs $\mathbf{C}(i, j)$. This range is determined directly from the distribution of segment costs in the given dataset. Minimizing this total cost function, thus corresponds directly to maximizing the posterior probability.

3) Optimal Segmentation via Dynamic Programming: Due to the additive structure of the total cost function, dynamic programming efficiently identifies the optimal segmentation. Let $\Gamma(j)$ be the minimal total cost for optimally segmenting the first j spectral bands, initialized by $\Gamma(0) = 0$. The recursion relation is defined as

$$\Gamma(j) = \min_{i=1,\ldots,j} \{\Gamma(i-1) + \mathbf{C}(i, j) + \lambda\}. \tag{13}$$

To further improve computational efficiency, the PELT algorithm [46] is integrated. PELT reduces computational overhead by iteratively discarding segmentation points that cannot yield optimal solutions. In particular, a candidate segmentation point τ is pruned if there exists a subsequent candidate $s > \tau$ satisfying

$$\Gamma(\tau) + \mathbf{C}(\tau + 1 : s) + \lambda \ge \Gamma(s) \tag{14}$$

then τ becomes redundant for all future time steps $t \ge s$. This follows because the cumulative cost of retaining τ as a candidate (i.e., $\Gamma(\tau) + C(\tau + 1 : t) + \beta$) will always be dominated by the alternative path through s, which satisfies

$$\Gamma(\tau) + \mathbf{C}(\tau + 1 : t) + \beta \ge \Gamma(s) + \mathbf{C}(s + 1 : t) + \lambda. \tag{15}$$

The pruning step does not compromise the optimality of the final segmentation, as the pruned candidates are provably suboptimal under all circumstances. Through successive pruning, PELT significantly reduces the number of candidates to evaluate. Under typical conditions observed in real-world data, the average size of the candidate set remains constant, yielding an overall time complexity of $O(n)$, which is substantially faster than unpruned optimal partitioning algorithms.

Algorithm 1 Optimal Band Segmentation With PELT

Input: Number of bands L, cost matrix $\mathbf{C} \in \mathbb{R}^{L\times L}$, penalty λ
Output: Change points $\mathcal{T} = \{\tau_k\}_{k=1}^{K}$, minimal cost $\mathcal{C}_{Total}$

Initialize $\boldsymbol{dp}(0:L) \leftarrow \infty$ ▷0-based indexing
$\boldsymbol{dp}(0) \leftarrow 0$ ▷Initial state: no cost before position 0
Initialize $\mathbf{B}_{tra}(0:L) \leftarrow 0$ ▷Backtracking pointer
$\mathcal{S}_c \leftarrow \{0\}$ ▷Initial candidate set
for $j = 1$ **to** L **do** ▷Iterate through each position
  $minCost \leftarrow \infty$, $bestTau \leftarrow 0$
  for $i \in \mathcal{S}_c$ **do**
    if $i \ge j$ **then continue**
    end if▷Skip invalid candidates
    $cost \leftarrow \boldsymbol{dp}(i) + \mathbf{C}(i+1, j) + \lambda$
    if $cost < minCost$ **then**
      $minCost \leftarrow cost$
      $bestTau \leftarrow i$ ▷Record optimal predecessor
    end if
  end for
  $\boldsymbol{dp}(j) \leftarrow minCost$
  $\mathbf{B}_{tra}(j) \leftarrow bestTau$
  $\mathcal{S}_{\text{prune}} \leftarrow \emptyset$
  for $i \in \mathcal{S}_c$ **do** ▷Check all candidates
    if $\boldsymbol{dp}(i) + \mathbf{C}(i+1, j) \ge \boldsymbol{dp}(j)$ **then**
      $\mathcal{S}_{\text{prune}} \leftarrow \mathcal{S}_{\text{prune}} \cup \{i\}$
    end if
  end for
  $\mathcal{S}_c \leftarrow \mathcal{S}_c \setminus \mathcal{S}_{\text{prune}}$
  $\mathcal{S}_c \leftarrow \mathcal{S}_c \cup \{j\}$
end for
$\mathcal{T} \leftarrow \emptyset$, $current \leftarrow L$
while $current > 0$ **do**
  $prev \leftarrow \mathbf{B}_{tra}(current)$
  if $prev \ge 0$ **then**
    $\mathcal{T} \leftarrow \mathcal{T} \cup \{prev\}$ ▷Record change point
  end if
  $current \leftarrow prev$
end while
$\mathcal{T} \leftarrow \text{sort}(\mathcal{T})$
$\mathcal{C}_{Total} \leftarrow \boldsymbol{dp}(L)$
return $\mathcal{T}, \mathcal{C}_{Total}$

Algorithm 1 provides pseudocode illustrating the process. The cost matrix $\mathbf{C} \in \mathbb{R}^{L\times L}$ calculated from KDE is one of the inputs to the algorithm. Initialization establishes the dynamic-programming array $\boldsymbol{dp}$ and backtracking array $\mathbf{B}_{\text{tra}}$, where $\boldsymbol{dp}(0) = 0$ encodes the base case of zero segmentation cost for an empty sequence. The candidate set $\mathcal{S}_c$, initialized with the dummy index 0, dynamically tracks viable segmentation points. For each band j from 1 to L, the algorithm iterates over all candidates $i \in \mathcal{S}_c$, computing the tentative cost

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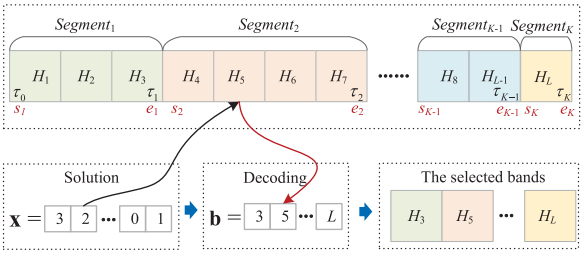


Fig. 2. Visualization of group integer encoding.

$\mathrm{dp}(i) + \mathbf{C}(i+1, j) + \lambda$. This quantifies the total penalty if a new segment $[i+1, j]$ is created, anchored by the optimal partitioning of bands 1 through i. The minimal cost across all candidates is assigned to $\boldsymbol{dp}(j)$, with $\mathbf{B}_{\mathrm{tra}}(j)$ recording the optimal split point i. Subsequently, PELT pruning eliminates candidates satisfying $\boldsymbol{dp}(i) + \mathbf{C}(i+1, j) \geq \boldsymbol{dp}(j)$, ensuring only indices capable of improving future costs are retained in $\mathcal{S}_c$. Once the forward pass completes, the algorithm recovers the change point set $\mathcal{T}$ through backtracking. Starting from band L, the recorded optimal predecessors in $\mathbf{B}_{\mathrm{tra}}$ are sequentially traced back to generate the final set of segmentation points $\{\tau_k\}_{k=1}^{K}$. The method returns this optimal segmentation and the corresponding minimal total cost.

C. Multiobjective Optimization for BS

1) Segmenting-Guided Integer Encoding and Decoding: The representative BS problem is formulated as a multiobjective optimization (MOO) framework. Unlike conventional binary encoding methods, which result in a high-dimensional search space due to representing each band individually, this study introduces a segment-based integer encoding strategy. In particular, following segmentation through Bayesian nonparametric modeling, the spectral dimension is partitioned into K contiguous segments. Each segment j is defined by its starting and ending indices, denoted as s_j and e_j, respectively, where $s_1 = 1$, $s_j = \tau_{j-1} + 1$ for $j = 2, \ldots, K$, and $e_j = \tau_j$. To efficiently encode BS within the jth segment, an integer decision variable is introduced

$$x_j \in \{0, 1, 2, \ldots, e_j - s_j + 1\}. \tag{16}$$

The global band index corresponding to the selected band in segment j is then defined as

$$b_j = \mathrm{sgn}\left(x_j\right) \cdot \left(s_j + x_j - 1\right) \tag{17}$$

where the sign function $\mathrm{sgn}(x_j)$ equals 1 if $x_j > 0$ and 0 otherwise. Thus, this formulation implies no BS from segment j when $x_j = 0$, and assigns the correct global band index when $x_j > 0$. Consequently, a solution $\mathbf{x} = [x_1, \ldots, x_K]$ corresponds to a selected subset of bands represented as $\mathbf{b} = [b_1, \ldots, b_{K_b}]$, where K_b denotes the total number of selected bands. The illustration of group integer encoding is shown in Fig. 2.

This segment-based integer encoding approach represents the selection of spectral bands within each predefined segment as a single integer variable, reducing the decision variable space from L in traditional binary formulations to K (with $K \ll L$), significantly enhancing computational efficiency.

2) Optimization Modeling for BS: The MOO framework simultaneously considers two complementary objectives: minimizing redundancy among selected bands and maximizing their collective information content.

Objective 1 (Redundancy Minimization): Redundancy among selected bands is quantified using mean mutual information (MMI), derived from information theory, as follows:

$$\mathrm{MMI}\left(\mathbf{x}\right) = \frac{1}{C_{K_b}^2} \sum_{i=1}^{K_b-1} \sum_{j=i+1}^{K_b} J\left(H\left(b_i\right); H\left(b_j\right)\right) \tag{18}$$

with mutual information $J(H(b_i); H(b_j))$ computed as

$$J\left(H\left(b_i\right); H\left(b_j\right)\right) = \sum_{u \in H(b_i)} \sum_{v \in H(b_j)} P\left(u, v\right) \log \frac{P\left(u, v\right)}{P\left(u\right) P\left(v\right)} \tag{19}$$

where $C_{K_b}^2$ represents the number of band pairs. $P(u, v)$ denotes the joint probability of pixel values u and v at corresponding locations in bands $H(b_i)$ and $H(b_j)$, where $P(u)$ and $P(v)$ denote the probabilities of pixel values u and v in bands $H(b_i)$ and $H(b_j)$, respectively. Thus, the redundancy minimization objective is defined as

$$\min_{\mathbf{x} \in \mathbb{Z}^{K \times K}} f_1\left(\mathbf{x}\right) = \mathrm{MMI}\left(\mathbf{x}\right). \tag{20}$$

Objective 2 (Information Maximization):

To quantify the information content of the selected bands, Shannon entropy is utilized. Furthermore, bands near wavelengths known to be sensitive to critical elements in zinc concentrate are assumed to carry additional discriminative information. Denoting the set of indices of known sensitive bands as

$$Z = \left[z_1, z_2, \ldots, z_{K_z}\right]. \tag{21}$$

First, the Shannon entropy of $H(b_j)$ is calculated as

$$E\left(H\left(b_j\right)\right) = -\sum_{u \in H(b_j)} P\left(v\right) \log P\left(v\right) \tag{22}$$

with $P(v)$ denoting the probability of pixel intensity v in $H(b_j)$. Then, the distance measure $d(H(b_j), H(Z))$ is defined as follows:

$$d\left(H\left(b_j\right), H\left(Z\right)\right) = \min_{z \in Z} \left\{ \mathrm{JSD}\left(P\left(H\left(b_j\right)\right), P\left(H\left(z\right)\right)\right) + \frac{|b_j - z|}{L} \right\}. \tag{23}$$

where $\mathrm{JSD}(P(H(b_j)), P(H(z)))$ denotes the Jensen–Shannon (JS) divergence, which quantifies the similarity between the distributions of $H(b_j)$ and $H(z)$. The definition of JS divergence is as follows:

$$\begin{aligned} \mathrm{JSD}\left(P\left(H\left(b_j\right)\right), P\left(H\left(z\right)\right)\right) &= \frac{1}{2} \sum_{v} P\left(v\right) \log_2 \frac{P\left(v\right)}{0.5\left(P\left(v\right) + P\left(\mu\right)\right)} \\ &\quad + \frac{1}{2} \sum_{\mu} P\left(\mu\right) \log_2 \frac{P(\mu}{0.5\left(P\left(u\right) + P\left(\mu\right)\right)} \end{aligned} \tag{24}$$

where $P(\mu)$ denotes the probability of pixel intensity μ in $H(z)$, and the logarithm is taken with base 2. This definition ensures that the JSD has a range of $[0, 1]$, providing a bounded

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and finite measure for comparing distributions. Both JSD and the symmetric information divergence (SID) [47] are well-established symmetric variants of the Kullback–Leibler (KL) divergence and have been widely applied. However, we chose JSD in this work is specifically motivated by its bounded range aligns well with the design of the formulation in (23). This function defines the distance $d(H(b_j), H(Z))$ as an aggregation of two distinct metrics: distributional similarity (measured by JSD) and wavelength proximity ($|b_j - z|/L$). The wavelength proximity term is inherently normalized to the interval $[0, 1]$, as it measures the proximity between the current band and the sensitive bands. To ensure a balanced aggregation of these two metrics and to avoid the issue of large values overwhelming smaller ones, it is critical to employ a divergence metric with a commensurate, bounded range. The JSD, with its fixed range of $[0, 1]$, perfectly matches the scale of the wavelength proximity term. This is in contrast to a metric like SID, whose range is $[0, \infty)$.

Since $E(H(b_j))$ and $d(H(b_j), H(Z))$ differ in units and scale, both terms are normalized to the interval $[0, 1]$ based on the maximum and minimum values within the current population at each step of the multiobjective optimization. The resulting normalized values, $E'(H(b_j))$ and $d'(H(b_j), H(Z))$, ensure that both metrics contribute comparably to the information criterion. The information content of band $H(b_j)$ is then defined as

$$I\left(H\left(b_j\right)\right) = E'\left(H\left(b_j\right)\right) - d'\left(H\left(b_j\right), H\left(Z\right)\right). \tag{25}$$

Consequently, information maximization is converted into a minimization formulation as

$$\min_{\mathbf{x}\in\mathbb{Z}^{K\times K}} f_2\left(\mathbf{x}\right) = -\sum_{j=1}^{K_b} I\left(H\left(b_j\right)\right). \tag{26}$$

Combining these objectives, the MOO problem is succinctly expressed as

$$\begin{aligned} \min_{\mathbf{x}} \quad & F\left(\mathbf{x}\right) = \left[f_1\left(\mathbf{x}\right), f_2\left(\mathbf{x}\right)\right]^{\mathrm{T}} \\ \text{s.t.} \quad & x_j \in \left\{0, 1, \ldots, e_j - s_j + 1\right\}, \quad j = 1, \ldots, K. \end{aligned} \tag{27}$$

Finally, the improved version of the SPEA-II [48] is adopted to efficiently search for Pareto-optimal solutions in the reduced integer decision space, yielding optimal band subsets that achieve a balance between minimal redundancy and maximal information content. To select the most balanced solution from the Pareto front, the technique for order preference by similarity to an ideal solution (TOPSIS) [49], [50] is applied.

D. Computational Complexity of BNS-MOP

The dynamic programming with the PELT algorithm for optimal segmentation has a time complexity of $O(L)$, where L is the number of spectral bands. The multiobjective optimization step using SPEA-II has a time complexity of $O(\mathrm{GP}^2M)$, where G is the number of generations, P is the population size, and M is the number of objectives. Therefore, the total complexity can be expressed as $O(\mathrm{GP}^2M)$.

TABLE I
DESCRIPTIVE STATISTICS OF ELEMENT CONCENTRATIONS IN ZINC CONCENTRATE SAMPLES

Statistic	Zn (%)	Fe (%)	Pb (%)	Cu (%)
Max	88.8500	21.7100	6.0900	2.2600
Min	73.5000	6.9400	1.9000	0.4400
Mean	81.7206	11.8725	3.5960	1.1272
Std	4.0883	4.0118	1.1219	0.3854
Skewness	-0.0639	0.9036	0.5818	0.8041
Kurtosis	2.0161	2.8073	2.4571	4.2309

IV. EXPERIMENTAL RESULTS AND ANALYSIS

A. Dataset Description and Data Preprocessing

1) Dataset Description: The dataset used in this study consists of hyperspectral measurements from 625 powdered zinc-concentrate samples. Table I summarizes the descriptive statistics of elemental concentrations. Each sample was measured five times to mitigate random fluctuations such as dark current noise and improve spectral reliability. The original hyperspectral data contained 512 bands, spanning wavelengths from 755 to 1730 nm. However, the initial 92 bands (755–932 nm) exhibited significant noise due to optical limitations of the instrument and were therefore excluded. The remaining 420 bands, characterized by higher signal-to-noise ratios, were retained. Each hyperspectral image simultaneously captures multiple physical samples arranged within the field of view. Therefore, spatial segmentation was applied to isolate individual sample regions within each hyperspectral image, resulting in band images of 20×20 pixels per sample.

2) Data Preprocessing: Initially, white and dark reference calibrations were applied to convert raw spectral intensities into standardized reflectance values, eliminating instrument response and ambient illumination effects. Repeated measurements for each sample were averaged to obtain a representative spectrum. Subsequently, spectra were smoothed using Savitzky–Golay filtering [51] to enhance signal quality. Baseline correction, conducted through low-order polynomial fitting, addressed scattering-induced spectral variations. To further mitigate residual scattering and amplitude variability, the standard normal variate (SNV) transformation [52] was applied. Finally, all spectra were normalized to a common scale, ensuring consistency across samples. The fully preprocessed dataset $\mathbf{H} \in \mathbb{R}^{20\times20\times420}$ served as input for subsequent analyses.

3) Element-Specific Sensitive Wavelength Selection: To obtain element-specific sensitive wavelengths for Zn, Fe, Pb, and Cu, we used a PLS-variable importance in the projection (VIP) procedure [53], [54] that is widely applied in chemometrics for assessing variable relevance in high-dimensional spectral data. For each element, a 1-D PLS regression model was fit using the preprocessed full-spectrum reflectance (420 bands) as predictors and the elemental concentrations as responses, and the optimal number of latent variables A_{opt} was selected by ten-fold cross-validation over $A = 1, \ldots, 15$. VIP scores were then computed for all wavelengths, and the top 10% were retained as element-specific sensitive wavelengths. PLS–VIP is used here as an

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interpretable wavelength-importance analysis, while RF and convolutional neural network (CNN) models are subsequently employed to evaluate the predictive value of these band subsets. The complete sensitive-wavelength intervals and their graphical illustrations are reported in Supplementary Table S1 and Supplementary Fig. S1.

B. Experimental Settings

1) Comparative Band-Selection Algorithms: To evaluate the effectiveness of the proposed BNS-MOP method, comparisons were carried out against ten state-of-the-art hyperspectral BS algorithms. These include the ranking-based approach: SMI-BS [14], the ranking-based clustering approaches: b_c-BDPC [31] and E-FDPC [30], the graph self-representation-based approach: TGSR [55], as well as four clustering-based methods: SLFLRSC [56], FNGBS [57], OCF [23], NGNMF [15], and two optimization-based methods: ONR [11] and MOCS-BS [58], with the latter formulated as a multiobjective-optimization-based approach. Parameters for each comparative method were configured according to the recommended settings provided in their respective original publications to ensure optimal performance.

2) Predictive Modeling With RF: To map the selected spectral bands to elemental concentrations, we adopt a random forest (RF) regressor as the main predictive model. Our focus is on the behavior of different band-selection strategies rather than on designing a sophisticated regression architecture.

a) RF regressor: The RF model comprises 100 trees with bootstrap aggregation enabled, and the number of candidate features at each split is set to $(K)^{1/2}$, where K is the number of selected bands. RF training employs a fixed random seed, while all remaining hyperparameters follow the Scikit-learn defaults. This configuration is kept fixed across all experiments so that performance differences can be attributed to the selected band subsets.

To verify that the observed advantages of the proposed band-selection method are not tied to the RF architecture, we also repeat all experiments with a CNN regressor. A short description of the CNN and the complete CNN-based mse and R^2 metrics are provided in the Supplementary Table S2 and Table S3.

b) Repeated fivefold cross-validation: Both models were evaluated with two repeats of fivefold cross-validation; the data splits were generated deterministically with a seed of 123. This procedure results in ten independent experiments per method, reduces the influence of any single data partition, and ensures that all methods are assessed on identical training and testing splits for fair and reliable comparison.

C. Regression Performance Comparisons

To comprehensively evaluate the predictive utility of band subsets obtained by different selection algorithms, eight target subset sizes (10, 15, 20, 25, 30, 40, 50, and 60 bands) were examined. For each subset size, spectral bands were selected by the proposed BNS-MOP method and the five comparative algorithms. Performance was evaluated in predicting Zn, Fe, Pb, and Cu concentrations using mse and the coefficient of determination (R^2). Since the selected bands from each method are relatively stable for a fixed number of bands, the BS process is performed once per method per setting.

For each band subset, regression performance is summarized as the mean and standard deviation over ten independent experiments. Statistical significance was assessed using Wilcoxon rank-sum tests at the 0.05 significance level. Symbols "+," "–," and "=" are used to indicate whether a competing method performs significantly better, worse, or similar to BNS-MOP, respectively. The best-performing result for each element under each band count is highlighted in bold and gray, while other results that outperform or are statistically equivalent to BNS-MOP are highlighted in gray.

Table II presents a comparison of RF-based elemental concentration estimation performance between the proposed BNS-MOP and ten state-of-the-art band-selection algorithms. For Zn, SMI-BS achieves the lowest mse with ten selected bands. BNS-MOP attains the best performance for most settings, including 20–30 and 50–60 bands, while b_c-BDPC and SLFLRSC provide strong competing results. For Fe, BNS-MOP maintains the lowest mse from 10 to 25 bands and at 40 bands. At other bands' counts settings, SLFLRSC, b_c-BDPC, ONR, and MOCS-BS achieve mse that are statistically better than the optimal result of BNS-MOP. For Pb, BNS-MOP is optimal at 10, 20, 30, and 60 bands, while NGNMF and OCF obtain the lowest mse at 15 and 40 bands, respectively. The b_c-BDPC and SLFLRSC obtain the lowest mse at 25 and 50 bands, respectively. Regarding Cu prediction, SLFLRSC shows a clear advantage at ten bands. BNS-MOP achieves the minimum mse at 15 and 30 bands, and at 60 bands, b_c-BDPC again provides performance comparable to that of BNS-MOP. In brief, although several baselines such as b_c-BDPC and SLFLRSC perform strongly, the proposed BNS-MOP still attains lower or comparable mse for most elements and band numbers, which confirms its overall effectiveness.

Fig. 3 illustrates the mse of elemental concentration predictions for four elements using different band-selection methods and varying numbers of selected bands. Circles denote RF predictor results and diamonds denote CNN predictor results, with each color indicating a different band-selection algorithms. The black dashed horizontal line in each panel corresponds to the mse of the full-band RF model, and the magenta dashed line corresponds to the mse of the full-band CNN model. Overall, the RF predictor consistently achieves lower mse than the CNN across all elements and almost all band-selection settings. As the K increases, the mse typically decreases at first, reaches a minimum at an intermediate band count, and then either increases again or saturates, which indicates that using more bands does not necessarily lead to better performance. This behavior highlights the importance of selecting an appropriate number of informative bands in order to achieve the best predictive accuracy.

Table III summarizes the R^2 scores of the RF-based elemental concentration estimates. The patterns are highly consistent with the mse results in Table II. BNS-MOP generally attains the highest or statistically comparable R^2 values for most elements and band numbers, especially in the medium band range from 25 to 40 selected bands. Several baselines, such

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TABLE II

MSE OF RF-BASED ELEMENTAL CONCENTRATION ESTIMATION ACROSS VARYING BAND COUNTS: COMPARATIVE PERFORMANCE OF THE PROPOSED BNS-MOP VERSUS TEN STATE-OF-THE-ART HYPERSPECTRAL BAND-SELECTION ALGORITHMS

Element	Algorithm	10 Bands	15 Bands	20 Bands	25 Bands	30 Bands	40 Bands	50 Bands	60 Bands
Zn	b_c-BDPC	0.0244(0.0025)–	0.0206(0.0016)–	0.0172(0.0019)=	0.0190(0.0015)–	0.0165(0.0016)=	**0.0120(0.0047)+**	0.0151(0.0024)=	0.0162(0.0022)–
	SMI-BS	**0.0199(0.0015)+**	0.0207(0.0020)–	0.0165(0.0022)=	0.0198(0.0031)–	0.0176(0.0027)–	0.0163(0.0024)=	0.0154(0.0033)–	0.0166(0.0017)–
	SLFLRSC	0.0213(0.0023)=	0.0194(0.0029)–	0.0179(0.0018)–	0.0172(0.0023)=	0.0166(0.0034)=	0.0132(0.0027)+	0.0158(0.0044)–	0.0167(0.0017)–
	E-FDPC	0.0292(0.0041)–	0.0234(0.0044)–	0.0221(0.0040)–	0.0175(0.0030)–	0.0187(0.0017)–	0.0178(0.0032)–	0.0157(0.0022)=	0.0174(0.0039)–
	TGSR	0.0240(0.0030)–	0.0213(0.0030)–	0.0194(0.0019)–	0.0190(0.0040)–	0.0181(0.0022)–	0.0167(0.0025)–	0.0186(0.0024)–	0.0171(0.0025)–
	ONR	0.0230(0.0026)–	0.0209(0.0018)–	0.0199(0.0025)–	0.0187(0.0024)–	0.0188(0.0041)–	0.0182(0.0014)–	0.0176(0.0007)–	0.0159(0.0016)–
	FNGBS	0.0237(0.0037)–	0.0234(0.0042)–	0.0177(0.0008)=	0.0185(0.0023)–	0.0176(0.0027)–	0.0180(0.0012)–	0.0184(0.0043)–	0.0175(0.0023)–
	OCF	0.0246(0.0032)–	0.0205(0.0035)–	0.0180(0.0021)–	0.0179(0.0032)–	0.0166(0.0017)=	0.0133(0.0015)=	0.0190(0.0030)–	0.0170(0.0038)–
	NGNMF	0.0212(0.0016)=	0.0195(0.0017)=	0.0193(0.0028)–	0.0180(0.0019)–	0.0171(0.0015)–	0.0160(0.0011)–	0.0161(0.0016)–	0.0158(0.0026)–
	MOCS-BS	0.0219(0.0009)–	**0.0183(0.0022)+**	0.0209(0.0035)–	0.0184(0.0023)–	0.0179(0.0024)–	0.0175(0.0022)–	0.0155(0.0031)=	0.0159(0.0027)–
	BNS-MOP	0.0214(0.0040)	0.0190(0.0022)	**0.0164(0.0008)**	**0.0172(0.0013)**	**0.0163(0.0027)**	0.0165(0.0030)	**0.0149(0.0017)**	**0.0158(0.0011)**
Fe	b_c-BDPC	0.0212(0.0030)–	**0.0194(0.0079)=**	0.0218(0.0050)–	0.0176(0.0041)=	0.0164(0.0057)+	0.0199(0.0028)–	0.0193(0.0027)–	0.0185(0.0032)+
	SMI-BS	0.0206(0.0022)=	0.0206(0.0016)–	0.0194(0.0027)–	0.0192(0.0033)–	0.0183(0.0032)–	0.0182(0.0048)=	0.0199(0.0024)–	0.0187(0.0042)=
	SLFLRSC	0.0217(0.0050)–	0.0215(0.0021)–	0.0208(0.0026)–	0.0175(0.0024)=	**0.0157(0.0068)+**	0.0211(0.0055)–	0.0195(0.0074)–	0.0192(0.0027)–
	E-FDPC	0.0281(0.0030)–	0.0207(0.0015)–	0.0215(0.0024)–	0.0201(0.0018)–	0.0207(0.0046)–	0.0224(0.0048)–	0.0196(0.0044)–	0.0229(0.0039)–
	TGSR	0.0261(0.0022)–	0.0219(0.0035)–	0.0246(0.0014)–	0.0182(0.0026)=	0.0163(0.0030)+	0.0224(0.0051)–	0.0195(0.0049)–	0.0201(0.0034)–
	ONR	0.0246(0.0015)–	0.0200(0.0021)–	0.0237(0.0026)–	0.0213(0.0023)–	0.0220(0.0037)–	0.0213(0.0046)–	**0.0181(0.0042)+**	0.0229(0.0064)–
	FNGBS	0.0291(0.0039)–	0.0246(0.0023)–	0.0246(0.0042)–	0.0221(0.0039)–	0.0176(0.0036)=	0.0208(0.0037)–	0.0190(0.0063)=	0.0189(0.0038)–
	OCF	0.0261(0.0038)–	0.0219(0.0024)–	0.0215(0.0050)–	0.0208(0.0042)–	0.0185(0.0031)–	0.0195(0.0021)=	0.0196(0.0055)–	0.0211(0.0035)–
	NGNMF	0.0232(0.0034)–	0.0202(0.0038)–	0.0209(0.0036)–	0.0186(0.0018)–	0.0190(0.0044)–	0.0199(0.0038)–	0.0198(0.0030)–	0.0192(0.0036)–
	MOCS-BS	0.0228(0.0024)–	0.0213(0.0037)–	0.0240(0.0065)–	0.0206(0.0038)–	0.0186(0.0039)–	0.0201(0.0045)–	0.0213(0.0050)–	**0.0183(0.0023)+**
	BNS-MOP	**0.0203(0.0045)**	**0.0194(0.0024)**	**0.0193(0.0047)**	**0.0174(0.0039)**	0.0174(0.0022)	**0.0181(0.0043)**	0.0190(0.0035)	0.0187(0.0054)
Pb	b_c-BDPC	0.0227(0.0034)–	0.0139(0.0017)+	0.0139(0.0027)–	**0.0133(0.0029)=**	0.0137(0.0018)–	0.0144(0.0026)=	0.0133(0.0025)–	0.0124(0.0029)=
	SMI-BS	0.0170(0.0032)=	0.0133(0.0027)+	0.0143(0.0014)–	0.0139(0.0014)–	0.0136(0.0022)–	0.0149(0.0031)–	0.0136(0.0025)–	0.0137(0.0022)–
	SLFLRSC	0.0188(0.0019)=	0.0134(0.0023)+	0.0135(0.0038)–	0.0159(0.0015)	0.0134(0.0028)–	0.0153(0.0029)–	**0.0127(0.0015)=**	0.0129(0.0028)=
	E-FDPC	0.0254(0.0036)–	0.0185(0.0018)–	0.0192(0.0031)–	0.0160(0.0017)–	0.0152(0.0026)–	0.0178(0.0047)–	0.0140(0.0020)–	0.0145(0.0016)–
	TGSR	0.0226(0.0034)–	0.0175(0.0021)–	0.0206(0.0030)–	0.0184(0.0032)=	0.0140(0.0020)=	0.0156(0.0010)–	0.0131(0.0014)–	0.0145(0.0025)–
	ONR	0.0196(0.0021)–	0.0174(0.0024)–	0.0164(0.0032)–	0.0175(0.0030)=	0.0158(0.0020)–	0.0157(0.0049)–	0.0133(0.0019)–	0.0129(0.0018)–
	FNGBS	0.0224(0.0010)–	0.0209(0.0029)–	0.0157(0.0016)–	0.0152(0.0015)–	0.0157(0.0018)–	0.0173(0.0058)–	0.0143(0.0024)–	0.0134(0.0014)–
	OCF	0.0210(0.0030)–	0.0198(0.0050)–	0.0170(0.0012)–	0.0144(0.0022)–	0.0161(0.0010)–	**0.0125(0.0008)=**	0.0142(0.0037)–	0.0129(0.0020)–
	NGNMF	0.0188(0.0017)–	**0.0132(0.0013)+**	0.0178(0.0046)–	0.0137(0.0009)–	0.0145(0.0018)–	0.0151(0.0020)–	0.0145(0.0034)–	0.0132(0.0013)–
	MOCS-BS	0.0205(0.0037)–	0.0164(0.0047)–	0.0170(0.0019)–	0.0177(0.0021)–	0.0142(0.0016)–	0.0127(0.0020)+	0.0139(0.0017)–	0.0126(0.0019)=
	BNS-MOP	**0.0170(0.0023)**	0.0153(0.0023)	**0.0126(0.0013)**	0.0135(0.0008)	**0.0129(0.0015)**	0.0143(0.0024)	**0.0127(0.0023)**	**0.0122(0.0012)**
Cu	b_c-BDPC	0.0157(0.0008)–	0.0137(0.0036)–	0.0154(0.0016)–	0.0134(0.0027)–	0.0122(0.0049)=	0.0119(0.0023)=	0.0138(0.0025)–	**0.0124(0.0036)+**
	SMI-BS	0.0142(0.0029)=	0.0142(0.0016)–	**0.0129(0.0017)+**	0.0136(0.0006)–	0.0136(0.0027)–	0.0122(0.0019)–	0.0129(0.0018)=	0.0129(0.0018)–
	SLFLRSC	**0.0104(0.0039)+**	0.0133(0.0020)=	0.0135(0.0016)=	0.0127(0.0022)–	0.0125(0.0018)=	0.0120(0.0031)=	0.0132(0.0030)–	0.0125(0.0053)+
	E-FDPC	0.0196(0.0019)–	0.0144(0.0014)–	0.0135(0.0008)=	0.0137(0.0022)–	0.0135(0.0006)–	0.0129(0.0020)=	0.0150(0.0032)–	0.0135(0.0035)–
	TGSR	0.0198(0.0023)–	0.0166(0.0012)–	0.0162(0.0020)–	0.0135(0.0013)–	0.0126(0.0015)=	0.0125(0.0008)=	**0.0122(0.0014)+**	0.0134(0.0024)–
	ONR	0.0174(0.0019)–	0.0162(0.0025)–	0.0141(0.0028)=	0.0149(0.0046)–	0.0140(0.0032)–	0.0123(0.0020)=	0.0140(0.0030)–	0.0132(0.0035)–
	FNGBS	0.0185(0.0017)–	0.0154(0.0025)–	0.0152(0.0025)–	0.0136(0.0043)–	0.0136(0.0030)–	0.0165(0.0065)–	0.0123(0.0012)+	0.0130(0.0006)–
	OCF	0.0187(0.0018)–	0.0149(0.0022)–	0.0134(0.0022)+	**0.0117(0.0020)+**	0.0124(0.0025)–	0.0138(0.0027)–	0.0128(0.0025)+	0.0144(0.0021)–
	NGNMF	0.0167(0.0018)=	0.0136(0.0021)–	0.0143(0.0011)–	0.0126(0.0016)–	0.0131(0.0026)–	0.0130(0.0016)–	0.0138(0.0015)–	0.0129(0.0021)–
	MOCS-BS	0.0170(0.0036)–	0.0154(0.0028)–	0.0158(0.0035)–	0.0158(0.0013)–	0.0140(0.0020)–	**0.0106(0.0008)+**	0.0136(0.0029)–	0.0128(0.0022)–
	BNS-MOP	0.0144(0.0016)	**0.0128(0.0028)**	0.0139(0.0041)	0.0122(0.0020)	**0.0118(0.0020)**	0.0119(0.0010)	0.0129(0.0027)	0.0128(0.0039)
+/–/=		2/31/7	5/32/3	2/32/6	1/32/7	3/29/8	4/25/11	4/30/6	4/32/4

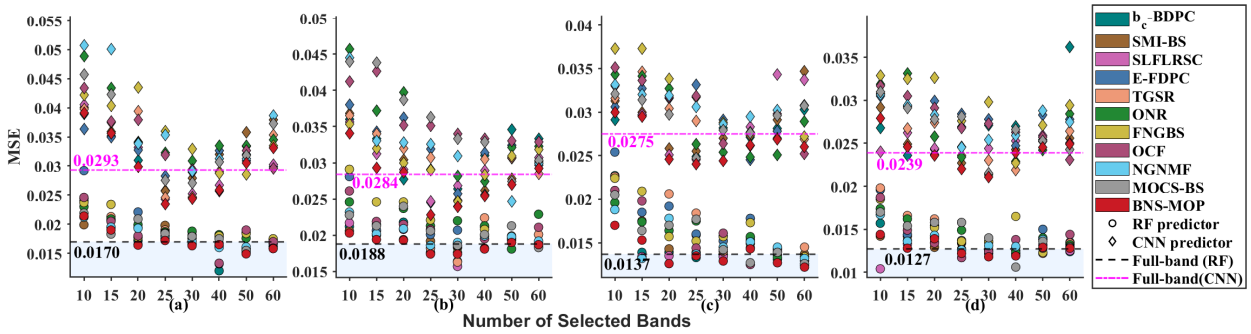


Fig. 3. MSE performance of RF and CNN models with varying band counts and selection approaches. (a) Zn, (b) Fe, (c) Pb, and (d) Cu.

as b_c-BDPC, SMI-BS, and SLFLRSC, can match or slightly exceed BNS-MOP for a few specific band settings, yet the summary row shows that BNS-MOP is rarely significantly worse and is more often significantly better than the competing methods.

Fig. 4 presents the R^2 values for predicting Zn, Fe, Pb, and Cu concentrations using different band-selection methods. Circles indicate RF predictions and diamonds represent CNN predictions, and each color corresponds to a different band-selection algorithms. The black dashed horizontal line in each subgraph shows the R^2 of the full-band RF model, and the magenta dashed line shows the R^2 of the full-band CNN model. RF consistently achieves higher R^2 values than CNN across all elements and band numbers. The CNN model shows relatively poor predictive accuracy, particularly for Cu, where R^2 values are much lower compared to the other elements. This poor performance can be attributed to the low mean concentration and limited variability of Cu in the dataset, as indicated in Table I. CNN, as a deep learning model, generally requires a large amount of diverse and informative data to learn effective feature representations. When the target element has a low concentration, the samples exhibit highly similar target values, resulting in only subtle spectral differences. This makes it difficult for the CNN to extract discriminative features,

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TABLE III

R^2 OF RF-BASED ELEMENTAL CONCENTRATION ESTIMATION ACROSS VARYING BAND COUNTS: COMPARATIVE PERFORMANCE OF THE PROPOSED BNS-MOP VERSUS TEN STATE-OF-THE-ART HYPERSPECTRAL BAND-SELECTION ALGORITHMS

Element	Algorithm	10 Bands	15 Bands	20 Bands	25 Bands	30 Bands	40 Bands	50 Bands	60 Bands
Zn	b_c-BDPC	0.7082(0.0391)−	0.6784(0.0254)−	0.7818(0.0212)=	0.7338(0.0240)−	0.7860(0.0271)−	**0.8248(0.0630)+**	0.7569(0.0333)−	0.7666(0.0353)−
	SMI-BS	**0.7521(0.0314)+**	0.6825(0.0205)−	0.7861(0.0244)=	0.7137(0.0479)−	0.7089(0.0281)−	0.7721(0.0290)−	0.7738(0.0475)−	0.7417(0.0265)−
	SLFLRSC	0.7201(0.0297)=	0.6893(0.0321)−	0.7708(0.0243)−	0.7578(0.0322)−	0.7705(0.0444)=	0.8201(0.0287)+	0.7742(0.0369)−	0.7608(0.0214)−
	E-FDPC	0.6452(0.0407)−	0.7112(0.0583)−	0.7112(0.0517)−	0.7720(0.0429)=	0.7481(0.0228)−	0.7735(0.0340)−	0.7917(0.0401)=	0.7731(0.0551)−
	TGSR	0.7068(0.0314)−	0.7311(0.0308)−	0.7305(0.0206)−	0.7527(0.0504)−	0.7681(0.0276)−	0.7728(0.0410)−	0.7630(0.0338)−	0.7768(0.0289)−
	ONR	0.7132(0.0314)−	0.7317(0.0313)−	0.7446(0.0360)−	0.7561(0.0215)−	0.7478(0.0510)−	0.7656(0.0102)−	0.7731(0.0080)−	0.7880(0.0283)−
	FNGBS	0.7042(0.0463)−	0.6948(0.0482)−	0.7680(0.0071)−	0.7639(0.0227)−	0.7668(0.0354)−	0.7613(0.0232)−	0.7686(0.0557)−	0.7706(0.0310)−
	OCF	0.7045(0.0435)=	0.7507(0.0423)−	0.7576(0.0225)−	0.7729(0.0319)=	0.7844(0.0116)−	0.8238(0.0189)+	0.7472(0.0499)−	0.7783(0.0423)−
	NGNMF	0.7470(0.0140)+	**0.7746(0.0339)=**	0.7569(0.0339)−	0.7719(0.0280)−	0.7851(0.0162)−	0.7914(0.0202)−	0.7949(0.0199)−	0.7951(0.0302)−
	MOCS-BS	0.7300(0.0157)−	0.7586(0.0276)=	0.7549(0.0438)−	0.7703(0.0369)−	0.7914(0.0287)=	0.7765(0.0255)−	0.7985(0.0304)=	0.7953(0.0308)−
	BNS-MOP	0.7173(0.0522)	0.7536(0.0184)	**0.7880(0.0121)**	**0.7778(0.0191)**	**0.7915(0.0296)**	0.7929(0.0337)	**0.8054(0.0198)**	**0.7973(0.0110)**
Fe	b_c-BDPC	0.7290(0.0269)−	0.7352(0.0883)−	0.6933(0.0521)−	0.7703(0.0453)=	0.7806(0.0467)+	0.7173(0.0294)−	0.7221(0.0386)−	**0.7640(0.0332)+**
	SMI-BS	**0.7384(0.0148)+**	**0.7580(0.0236)+**	0.7171(0.0374)−	0.7666(0.0353)−	0.7180(0.0474)−	0.7401(0.0547)−	0.7160(0.0340)−	0.7639(0.0450)+
	SLFLRSC	0.7240(0.0528)	0.7545(0.0208)=	0.7205(0.0279)−	0.7196(0.0388)−	0.7842(0.0654)+	0.7141(0.0691)−	0.7320(0.0866)−	0.7561(0.0327)=
	E-FDPC	0.6604(0.0404)−	0.7352(0.0216)−	0.7207(0.0342)−	0.7399(0.0203)−	0.7181(0.0589)−	0.7261(0.0393)−	0.7400(0.0569)−	0.7136(0.0627)−
	TGSR	0.6716(0.0156)−	0.7086(0.0472)−	0.6901(0.0269)−	0.7598(0.0214)−	**0.7888(0.0329)+**	0.7142(0.0627)−	0.7417(0.0531)−	0.7403(0.0366)−
	ONR	0.7015(0.0174)−	0.7411(0.0243)−	0.7027(0.0366)−	0.7012(0.0506)−	0.7268(0.0279)−	0.7376(0.0506)−	**0.7672(0.0493)+**	0.7307(0.0679)−
	FNGBS	0.6476(0.0516)−	0.6638(0.0448)−	0.6910(0.0387)−	0.7314(0.0318)−	0.7696(0.0256)=	0.7322(0.0386)−	0.7670(0.0477)+	0.7474(0.0433)−
	OCF	0.6759(0.0446)−	0.7317(0.0314)−	0.7197(0.0594)−	0.7303(0.0455)−	0.7685(0.0291)−	0.7598(0.0266)=	0.7407(0.0736)−	0.7370(0.0404)−
	NGNMF	0.7222(0.0305)−	0.7413(0.0459)−	0.7306(0.0349)−	0.7580(0.0265)−	0.7571(0.0476)−	0.7428(0.0426)−	0.7473(0.0354)−	0.7577(0.0465)−
	MOCS-BS	0.7264(0.0226)−	0.7398(0.0346)−	0.7093(0.1010)−	0.7683(0.0499)−	0.7744(0.0419)+	0.7506(0.0278)−	0.7217(0.0524)−	0.7634(0.0208)+
	BNS-MOP	0.7360(0.0478)	0.7421(0.0401)	**0.7520(0.0531)**	**0.7823(0.0313)**	0.7705(0.0106)	**0.7762(0.0521)**	0.7560(0.0434)	0.7622(0.0447)
Pb	b_c-BDPC	0.7289(0.0297)−	0.7748(0.0286)=	0.7154(0.0333)−	0.7943(0.0431)=	0.7974(0.0205)−	0.8040(0.0184)=	0.8322(0.0303)=	0.8198(0.0291)−
	SMI-BS	**0.8055(0.0363)+**	0.7775(0.0373)=	0.7149(0.0261)−	0.7984(0.0240)=	0.8054(0.0322)−	0.7921(0.0470)−	0.8212(0.0286)−	0.8085(0.0181)−
	SLFLRSC	0.7778(0.0205)+	0.7794(0.0298)=	0.8156(0.0376)−	0.7682(0.0144)−	0.8167(0.0189)−	0.7975(0.0436)=	0.8328(0.0285)=	0.8166(0.0442)−
	E-FDPC	0.6738(0.0466)−	0.7614(0.0211)−	0.7447(0.0422)−	0.7804(0.0368)−	0.7994(0.0372)−	0.7769(0.0491)−	0.8112(0.0282)−	0.7969(0.0223)=
	TGSR	0.7269(0.0394)−	0.7674(0.0361)−	0.7101(0.0502)−	0.7725(0.0326)−	0.8174(0.0290)−	0.7890(0.0166)−	0.8321(0.0221)−	0.8188(0.0286)−
	ONR	0.7587(0.0341)−	0.7737(0.0285)−	0.7949(0.0302)−	0.7853(0.0249)−	0.7969(0.0238)−	0.7948(0.0720)−	0.8325(0.0195)−	0.8318(0.0304)−
	FNGBS	0.7044(0.0173)−	0.7336(0.0431)−	0.7927(0.0156)−	0.8088(0.0204)−	0.7959(0.0209)−	0.7852(0.0613)−	0.8225(0.0326)−	0.8223(0.0329)−
	OCF	0.7418(0.0285)−	0.7540(0.0563)−	0.7754(0.0227)−	0.8129(0.0248)−	0.7970(0.0142)−	**0.8390(0.0109)+**	0.8234(0.0578)−	0.8324(0.0218)−
	NGNMF	0.7786(0.0193)=	**0.8264(0.0183)=**	0.7657(0.0653)−	0.8273(0.0143)−	0.8159(0.0196)−	0.8018(0.0240)−	0.8143(0.0411)=	0.8316(0.0134)−
	MOCS-BS	0.7582(0.0534)−	0.8221(0.0640)=	0.8030(0.0237)−	0.8131(0.0294)−	0.8313(0.0255)=	0.8362(0.0223)+	0.8179(0.0266)−	0.8304(0.0271)=
	BNS-MOP	0.7739(0.0319)	0.7987(0.0303)	**0.8424(0.0218)**	**0.8307(0.0139)**	**0.8324(0.0223)**	0.8160(0.0220)	**0.8349(0.0358)**	**0.8335(0.0241)**
Cu	b_c-BDPC	0.6465(0.0330)−	0.7303(0.0402)−	0.6689(0.0360)−	0.7233(0.0368)−	0.7388(0.0567)−	0.7866(0.0445)=	0.7763(0.0510)=	0.7586(0.0630)=
	SMI-BS	0.6635(0.0608)−	0.7237(0.0370)−	0.7450(0.0393)=	0.7256(0.0135)−	0.7633(0.0450)−	0.7405(0.0429)−	0.7765(0.0410)=	0.7487(0.0221)−
	SLFLRSC	**0.7961(0.0458)+**	0.7445(0.0334)=	0.7077(0.0367)−	0.7520(0.0415)−	**0.7944(0.0365)+**	0.7689(0.0400)−	0.7923(0.0462)=	0.7332(0.0988)−
	E-FDPC	0.6858(0.0370)−	0.7515(0.0144)−	0.7466(0.0206)−	0.7616(0.0256)−	0.7556(0.0244)−	0.7611(0.0290)−	0.7304(0.0610)−	**0.7707(0.0518)+**
	TGSR	0.6913(0.0169)−	0.7210(0.0230)−	0.7169(0.0250)−	0.7478(0.0219)−	0.7752(0.0244)−	0.7792(0.0201)−	0.7845(0.0368)+	0.7695(0.0280)−
	ONR	0.7286(0.0190)−	0.7258(0.0346)−	0.7359(0.0478)−	0.7406(0.0651)−	0.7462(0.0445)−	0.7861(0.0308)−	0.7665(0.0353)+	0.7512(0.0530)−
	FNGBS	0.6913(0.0316)−	0.7429(0.0439)−	0.7419(0.0328)−	0.7555(0.0746)−	0.7639(0.0393)−	0.7236(0.0882)−	**0.7960(0.0192)=**	0.7602(0.0218)−
	OCF	0.6853(0.0617)−	0.7459(0.0444)−	0.7621(0.0391)+	0.7832(0.0349)=	0.7764(0.0386)−	0.7604(0.0404)−	0.7715(0.0314)−	0.7491(0.0347)−
	NGNMF	0.7429(0.0416)=	0.7537(0.0251)−	0.7537(0.0259)−	0.7710(0.0363)−	0.7882(0.0323)=	0.7647(0.0406)−	0.7584(0.0136)−	0.7694(0.0306)=
	MOCS-BS	0.7419(0.0653)=	0.7559(0.0635)−	**0.7755(0.0525)+**	0.7664(0.0381)−	0.7877(0.0425)=	**0.8192(0.0084)=**	0.7764(0.0244)=	0.7604(0.0377)−
	BNS-MOP	0.7383(0.0306)	**0.7570(0.0510)**	0.7557(0.0644)	**0.7887(0.0347)**	0.7791(0.0448)	0.7956(0.0238)	0.7775(0.0435)	0.7648(0.0628)
+/−/=		6/29/5	1/30/9	2/35/3	0/34/6	5/29/6	5/30/5	4/26/10	4/31/5

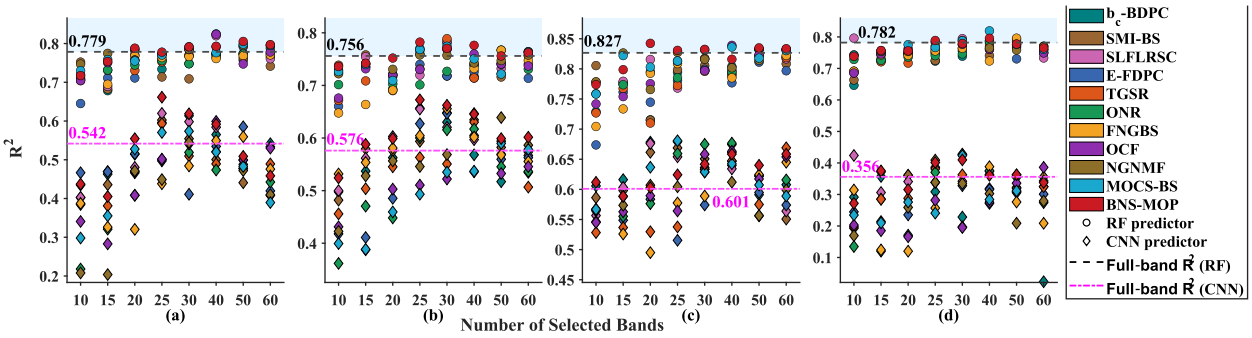


Fig. 4. R^2 performance of RF and CNN models with varying band counts and selection approaches. (a) Zn, (b) Fe, (c) Pb, and (d) Cu.

leading to reduced prediction accuracy. In contrast, the RF achieves substantially higher R^2 on Cu, as well as on other elements. This can be attributed to the tree-based model's ability to exploit a small number of highly informative spectral features and its robustness to label skewness, outliers, and noise. As a result, the RF provides more reliable regression performance, particularly for targets with narrow dynamic ranges and sparse predictive signals.

The computational runtime of each BS method is summarized in Fig. 5. Among the comparison algorithms, b_c-BDPC, E-FDPC, FNGBS, and OCF exhibit the highest computational efficiency, with runtimes remaining negligible (below 0.3 s) even as the number of selected bands increases. NGNMF and SMI-BS also maintain stable and relatively low runtimes, showing only a marginal increase compared to the fastest group. In contrast, SLFLRSC, TGSR, and ONR require noticeably more computation time. In particular, SLFLRSC and TGSR show a clear correlation between runtime and band count, which can be attributed to their iterative optimization or graph learning processes. Notably, MOCS-BS and the proposed BNS-MOP incur the highest computational costs among all methods, especially at larger band numbers. This increased

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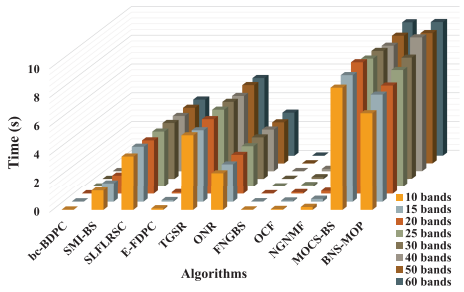


Fig. 5. Running time (s) comparison with varying band counts and selection approaches.

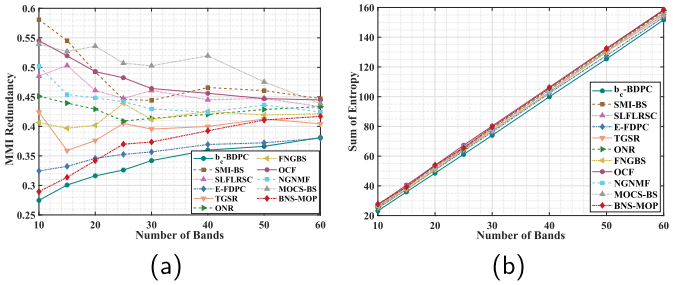


Fig. 6. Quantitative analysis across spectral band counts by different band-selection methods. (a) Redundancy. (b) Information content.

runtime is primarily attributable to the use of multiobjective optimization, where nondominated sorting and population-based search introduce additional computational overhead to ensure solution diversity and quality.

Overall, both mse and R^2 analyses confirm the superiority of BNS-MOP in selecting highly informative bands, with BNS-MOP consistently delivering superior predictive accuracy and stability across Zn, Fe, Pb, and Cu concentration estimations. One contributing factor is that, during BS, BNS-MOP evaluates information content using a criterion that combines conventional Shannon entropy with domain knowledge of element-sensitive-wavelength intervals. However, this advantage in predictive performance comes at the cost of higher computational time compared to most other methods, mainly due to the complexity of its MOO framework.

D. Analysis of The Selected Bands

To evaluate the quality of the bands selected by BNS-MOP, we analyze the statistical properties of the obtained band subsets.

In Fig. 6(a), the MMI curve of BNS-MOP increases almost monotonically with the number of bands. This behavior is similar to that of b_c-BDPC and E-FDPC and differs from the gently declining or nearly flat trajectories observed for most of the remaining algorithms. The rising trend reflects a strategy that first selects the most decorrelated bands. Once these have been exhausted, additional bands inevitably overlap spectrally with at least one already selected band, and the average redundancy therefore increases. The steadily increasing MMI of b_c-BDPC and E-FDPC likewise indicates that these methods also place emphasis on decorrelation in the early stages, followed by an unavoidable accumulation of redundancy as

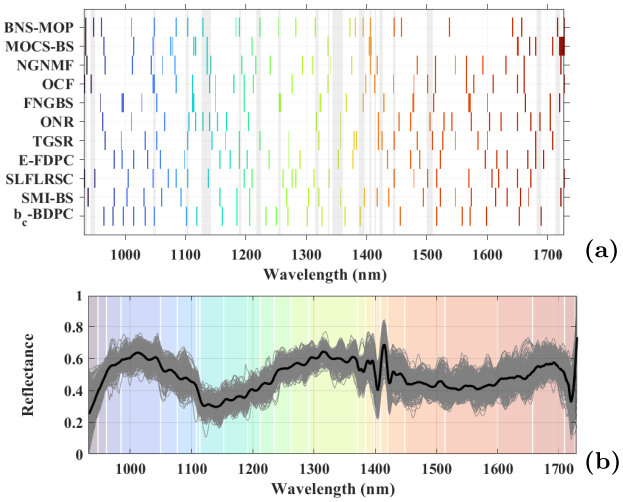


Fig. 7. Integrated visualization of spectral segmentation and band-selection results for 11 algorithms (illustrative example with 25 selected bands). (a) Locations of the bands selected by each algorithm. (b) Scene-averaged reflectance spectrum with 25 segments.

the subset grows. In contrast, methods whose MMI curves start at higher values and then remain almost flat tend to prioritize bands with strong individual information content. High-entropy bands often cluster in similar spectral regions, so these methods exhibit relatively high redundancy at small band counts, and this redundancy is later diluted as more diverse bands are added.

In Fig. 6(b), the sum of entropies increases approximately linearly with the number of bands for all methods, which shows that each additional band contributes new variability to some extent. The curve of BNS-MOP lies slightly above those of the competing approaches over most band counts, suggesting that it retains a large amount of information while still controlling redundancy. Taken together, these trends indicate that BNS-MOP achieves a favorable balance between redundancy reduction and information preservation across different band numbers.

Fig. 7 visualizes the spectral segmentation and band-selection results of the 11 algorithms for a representative scene with $K = 25$. In Fig. 7(a), each colored tick denotes a selected band and each row corresponds to one band-selection method. The light gray vertical strips indicate the element-specific sensitive-wavelength intervals derived from the PLS–VIP analysis described in Section IV-A3. The proposed BNS-MOP produces a dispersed selection pattern that covers the entire 932–1730-nm range and places a large fraction of its bands inside the sensitive regions, while also avoiding the dense local clustering that appears in several competing methods. The specific bands selected by 11 algorithms in this example, together with their information entropies, are listed in Supplementary Table S4.

In Fig. 7(b), the background shading shows the 25 contiguous segments obtained by the Bayesian nonparametric partitioning stage, and the gray strips again mark the sensitive-wavelength intervals. The scene-averaged reflectance curve (black line) reveals that many segment boundaries coincide with inflection points or pronounced absorption features,

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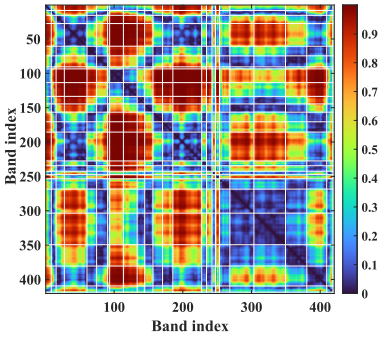


Fig. 8. Pairwise JS divergence matrix of the hyperspectral bands with Bayesian segment boundaries.

which supports the physical plausibility of the segmentation. By assigning at most one representative band to each homogeneous segment, BNS-MOP avoids repeatedly selecting highly correlated neighbors and encourages a compact yet informative coverage of both sensitive and nonsensitive regions.

These results jointly demonstrate the practical benefit of coupling similarity-based segmentation with multiobjective optimization, to be specific, by assigning at most one representative band to each homogeneous segment, BNS-MOP prevents the repeated selection of highly correlated neighbors. Consequently, BNS-MOP identifies subsets that not only maintain high information content but also demonstrate significantly reduced redundancy, particularly evident at intermediate subset sizes.

Fig. 8 shows the 420×420 heat map of pairwise JS divergences between spectral bands. For every band, we first estimate its pixel-value distribution with a kernel density estimator and then compute the JS divergence between every pair. JS divergence equals 0 for identical densities and increases monotonically with dissimilarity, so dark blue denotes high similarity, whereas yellow–red denotes marked contrast. The superimposed 25×25 white grid corresponds to the Bayesian change points reported in Fig. 7(b). Within each grid cell (i.e., inside a segment), the colors remain uniformly dark blue, confirming that bands assigned to the same segment exhibit statistically similar empirical distributions; across cells, the palette shifts sharply to green–red, signaling pronounced distributional changes at every learned boundary. Hence, the data themselves validate the modeling premise of segmentwise homogeneity with an unknown parametric form.

E. Impact of Band Count and Determination of the Unified Subset Size

This section investigates how the number of selected bands K influences prediction accuracy and selection stability, and provides a unified band count for practical use. In addition, an analysis of the impact of band count on the training dynamics and validation loss convergence of the underlying neural network is provided in the Supplementary Fig. S2.

1) Predictive Performance Across Various Band Counts: We evaluate predictive performance with respect to the number of selected bands K. For each K, ten independent BSs are generated. One model is trained for each selection, and the resulting mse values are summarized by boxplots for the four target elements.

The results in Fig. 9 reveal a coherent pattern across elements. Moving from very small band budgets to moderate band budgets yields a substantial reduction in mse. For Zn, the lowest mse occurs around K in the range 30–35. For Fe, the minimum lies near K from 20 to 30. For Pb, the mse decreases up to approximately $K = 45$ and then remains nearly unchanged. For Cu, the mse decreases steadily up to about $K = 45$ with only marginal variation beyond that point. These behaviors indicate that increasing K initially introduces complementary information that improves prediction, whereas very large K contributes mainly redundant or noisy bands and provides limited additional gain.

The boxplots provide evidence on robustness. At small K, the mse across the ten selections exhibits large between-selection differences, as indicated by wider interquartile ranges and longer whiskers. In the intermediate range $K \in [25, 45]$, the boxplots are compact, and the medians vary little. At the largest K, a slight widening is visible, which is consistent with the effect of redundant bands diluting the informative signal.

2) Consistency of Selected Bands Across K: Fig. 10 shows for each band budget K the selection frequency of each wavelength across the $R = 10$ band-selection solutions. The gray vertical bands indicate the element-specific sensitive-wavelength intervals derived from the PLS–VIP analysis in Supplementary Table S1. Most bands are selected rarely, whereas several compact spectral regions exhibit consistently high selection frequencies, and these regions largely lie within or adjacent to the sensitive intervals. When K increases, the overall selection frequency rises slightly because more bands can be retained, but the locations of the high-frequency regions remain almost unchanged, indicating that similar spectral regions are repeatedly recovered across runs and band budgets.

Fig. 11 provides a quantitative assessment of stability. For each K, the violin plot shows the empirical distribution of pairwise Jaccard similarity between band subsets from different runs. The median similarity increases from about 0.43–0.50 at small K to roughly 0.52–0.54 for K between 25 and 45, and then decreases slightly to about 0.47–0.49 for larger K. The spread is widest at small K, narrowest for K between 25 and 45, and moderately wider again at the largest K. Thus, repeated runs are most consistent in the moderate band-count range.

The analyses across varying band counts show that predictive errors are lowest and boxplots are most compact for K between 25 and 45. The selection frequency maps and pairwise Jaccard distributions further indicate concentration on a stable subset of bands with strong agreement across selections, which supports stability with respect to K rather than uncontrolled randomness.

3) Determination of the Unified Band Count: Building upon the regression performance analysis, we next determine a single, unified number of selected bands K that is applicable to all four target elements. To this end, we consider the same discrete set of candidate band budgets as in the comparative experiments, $\mathcal{K}_{\text{cand}} = \{10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60\}$, i.e., a grid with a step size of 5 bands. This choice provides

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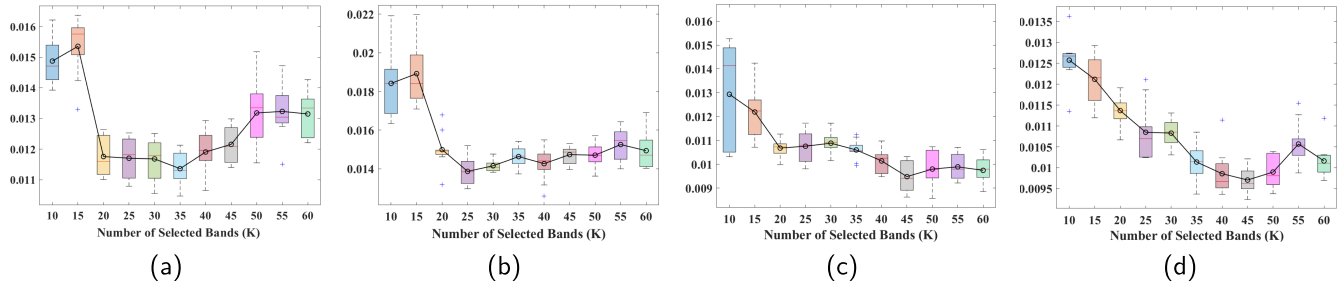


Fig. 9. MSE distributions across various band counts for four elements. (a) Zn. (b) Fe. (c) Pb. (d) Cu.

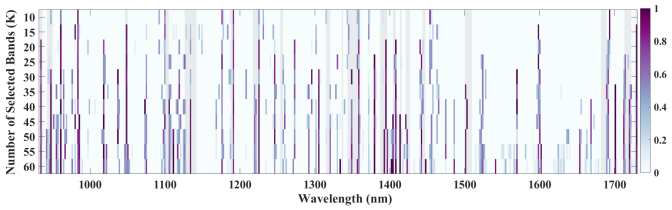


Fig. 10. Selection frequency heatmap across K.

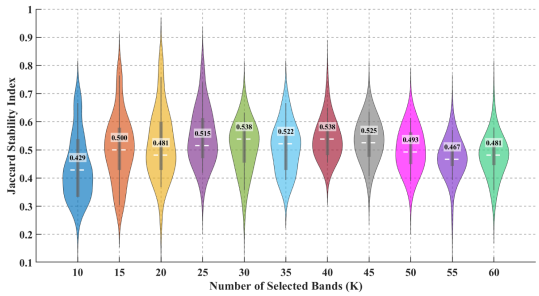


Fig. 11. Distribution of Jaccard similarity across K.

sufficient resolution while avoiding unnecessary computational overhead. Owing to the strong spectral correlation in hyperspectral data, adjacent bands carry highly redundant information, so moderate changes in K tend to induce only limited variations in the effective spectral content. Consequently, once K enters the near-optimal region, the marginal information gain from adding a single band is expected to be small. In this regime, using a unit step in K would rarely change the qualitative conclusions, and the grid with a step size of 5 bands is sufficiently dense to resolve the practically relevant operating range.

Each candidate $K \in \mathcal{K}_{\text{cand}}$ is evaluated according to two complementary criteria: 1) an overall relative prediction error $E_{\text{rel}}(K)$ that aggregates the performance across the four elements, and 2) the average Jaccard similarity $\bar{J}_K$, which quantifies the stability of the selected band subsets.

Let $\mathcal{E} = \{\text{Zn}, \text{Fe}, \text{Pb}, \text{Cu}\}$ be the set of target elements. For each $e \in \mathcal{E}$ and band budget K, we compute the average mean squared error $\overline{\text{mse}}_e(K)$ over repeated runs, and define $\text{mse}_{\min,e} = \min_{K \in \mathcal{K}_{\text{cand}}} \overline{\text{mse}}_e(K)$ as the best achieved performance for element e. The overall relative prediction error is then defined as

$$E_{\text{rel}}(K) = \frac{1}{|\mathcal{E}|} \sum_{e \in \mathcal{E}} \frac{\overline{\text{mse}}_e(K) - \text{mse}_{\min,e}}{\text{mse}_{\min,e}}. \tag{28}$$

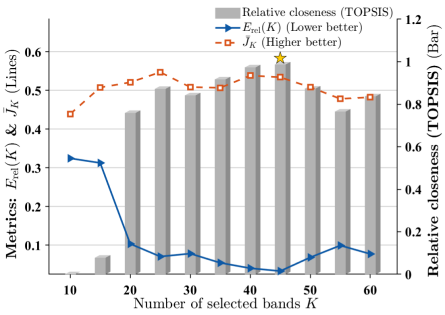


Fig. 12. Selection criteria and TOPSIS relative closeness coefficient for different K.

To characterize the stability of the selected bands, for each K BNS-MOP yields R band-subset solutions, denoted by $S^{(K,r)} \subset \{1, \ldots, L\}$ for $r = 1, \ldots, R$. The Jaccard similarity between two subsets with indices $r \neq s$ is given by

$$J\left(S^{(K,r)}, S^{(K,s)}\right) = \frac{\left|S^{(K,r)} \cap S^{(K,s)}\right|}{\left|S^{(K,r)} \cup S^{(K,s)}\right|} \tag{29}$$

and the average Jaccard similarity at band budget K is

$$\bar{J}_K = \frac{2}{R(R-1)} \sum_{1 \le r < s \le R} J\left(S^{(K,r)}, S^{(K,s)}\right). \tag{30}$$

Finally, treating $E_{\text{rel}}(K)$ as a cost-type criterion and $\bar{J}_K$ as a benefit-type criterion, we employ the TOPSIS method to rank the candidates in $\mathcal{K}_{\text{cand}}$. This procedure identifies $K^* = 45$ as the unified band budget. The behavior of $E_{\text{rel}}(K)$, $\bar{J}_K$, and the corresponding TOPSIS relative closeness coefficient is illustrated in Fig. 12.

V. Conclusion

This study proposed a Bayesian nonparametric segmentation integrated with a multiobjective optimization (BNS-MOP) approach to address the high dimensionality and redundancy of HSI for zinc concentrate compositional analysis. First, kernel density-based Bayesian segmentation with PELT partitions the spectrum into physically coherent segments while preserving spectral contiguity. On this basis, a multiobjective optimization model with segmentwise integer encoding is constructed to minimize interband redundancy and to maximize an information measure that combines conventional Shannon entropy with prior knowledge about element-specific sensitive-wavelength intervals. RF and CNN regression models are then

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used to predict elemental concentrations from the selected band subsets. Experiments on 645 real-world zinc-concentrate samples show that BNS-MOP consistently outperforms ten state-of-the-art band-selection methods, achieving lower mse and higher R^2 for all four elements. Beyond the comparison at fixed band numbers, we further analyzed how the total number of selected bands K affects both predictive performance and selection stability. For this method, RF-based prediction accuracy improves quickly as K increases from very small values and becomes stable once K reaches roughly 25–45, where the errors remain low, and the selected bands show high Jaccard stability. Balancing accuracy, stability, and model complexity, we therefore recommend a unified band budget of $K^* = 45$ for practical use.

Future work will focus on further enhancing the flexibility and spatial awareness of the proposed framework. On the density modeling side, it would be of interest to replace the current fixed bandwidth KDE with more adaptive schemes or variational Bayesian mixture models so that bandwise pixel distributions with multimodal or heavy-tailed behavior can be captured more accurately while controlling computational cost. Another promising direction is to incorporate spatial information into the band segmentation stage by enriching band-level descriptors with texture features or by defining band similarity measures that also reflect spatial structural agreement. Such spatial–spectral extensions are expected to strengthen the robustness of BNS-MOP in more complex imaging scenarios.

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